MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG1MDCPHY100 – Physics Around You

Duration: 1.5 hrs Maximum Marks: 35

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **all** questions

Each question carries **1** Mark

1 What is the unit of frequency? [U][1]

a) Newton b) Hertz c) Joule d) Pascal

2 Which of the following describes acceleration? [U][1]

a) Change in speed with time b) Change in velocity with time c) Constant velocity

d) Distance covered per second

3 What must be the angle between the applied force and displacement for doing maximum work? [U][1]

a) 0° b) 30° c) 45° d) 90°

4 Which of the following has 100 % efficiency? [U][1]

a) Electric motor b) Generator c) Real machine d) Ideal machine

5 Find kinetic energy of a 0.2 g insect when it is flying at 0.4m/s? [U][1]

Hint:Kinetic energy=$\frac{1}{2}mv^2$, where m is the mass and v is the velocity of the object.

a) 1.6×10^{-4}J b) 1.6×10^{-2}J c) 1.4 J d) 5.6×10^{-4}J

6 When particles in the medium vibrate parallel to the direction of propagation of a wave, the wave is called: [U][1]

a) Longitudinal wave b) Transverse wave c) Electromagnetic Wave d) radio waves

7 The loudness of sound is directly proportional to: [U][1]

a) Frequency b) wavelength c) Velocity d) Amplitude

8 The unit of gravitational constant G is: [U][1]

a) Nm^2/kg^2 b) kgm/s^2 c) m/s^2 d) Nm/s

9 What does a voltmeter measure? [K][2]

a) Current b) Resistance c) Voltage d) Power

10 Which of the following generates alternating voltage? [U][2]

a) Battery b) Generator c) Solar cell d) Resistor

11 The resistivity of a conductor depends on: [U][2]

a) Length b) Cross-sectional area c) Material and temperature d) Voltage

12 The power consumed by an electrical device is given by: [U][2]

a) $P = VI$ b) $P = (I^2)R$ c) $P = (V^2)/R$ d) All of the above

13 If the wavelength of electromagnetic radiation is doubled, what will happen to the energy. [U][3]

a) Doubled b) Halved c) Remains same d) Becomes zero

14 Which of the following is not transported by electromagnetic waves? [U][3]

a) Charge b) Information c) Energy d) Momentum

15 Which of the following is not a typical application of a laser? [K][4]

a) Cutting materials b) Data storage c) Thermal imaging d) Heating water

(1x15=15)

Part B

Multiple Choice Questions

Answer **all** questions

Each question carries **2** Mark

16 What is the dimensional formula for pressure? [U][1]

a) $ML^{-1}T^{-2}$ b) $M^0L^2T^{-3}$ c) ML^2T^{-1} d) ML^2T^{-2}

17 When a ball is thrown vertically upward, what is its acceleration during its motion? [U][1]

a) 0 m/s² b) 9.8 m/s² upward c) 9.8 m/s² downward d) Changes depending on the height

18 A 60 kg person climbs 5 m up a staircase. What is the work done? [U][1]
Hint: Work, W= F.S and Force F= ma
where S is the displacement and a is the acceleration.
Given acceleration due to gravity,g=9.8m/s^2
a) 2940 J b) 300 J c) 5880 J d) 1470 J

19 The speed of sound is: [U][1]

a) Slower in water than in air b) Faster in water than in air

c) Same in air and in water d) sound cannot travel in water

20 The Doppler Effect occurs when: [U][1]

a) A sound wave is reflected from a surface
b) A sound wave source is stationary
c) A sound wave source moves relative to an observer
d) The frequency of the wave decreases with time.

21 If the resistance is 10Ω and voltage is 50V, what is the current? [U][2]

Hint: Voltage, V=IR, I is the current and R is the resistance.

a)5A b)10A c)2A d)1A

22 What type of graph does Ohm's Law produce? [U][2]

a) Linear b) Exponential c) Parabolic d) Logarithmic

23 Waves in decreasing order of their wavelength are [K][3]

a) Radio waves, ultraviolet rays, visible light, X – rays
b) Radio waves, visible light, infrared rays, X- rays
c) Radio waves, infrared rays, visible light, X – Rays
d) X - rays, infrared rays, visible light, radio waves

24 If the angle of incidence is 30° and the refractive index is 1.33, calculate the angle of refraction using Snell's law. [A][3]

a) 15.67° b) 25.56° c) 22.09° d) 18.79°

25 The material in which population inversion can take place is called

a) active medium b) passive medium c) gaseous medium d) vapour medium [K][4]

(2x10=20)